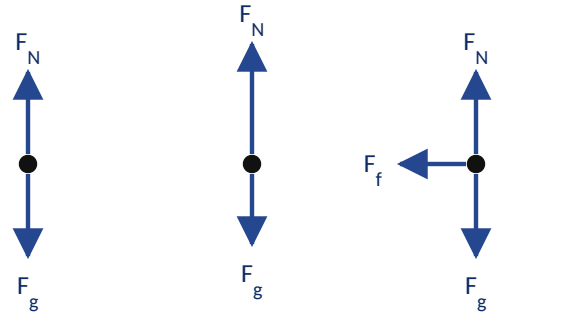
Introductory dynamics: cram sheet

AP Physics 1 · 50 min · 4 parts, 25% each

The test

- **Free-body diagram**: a dot, original forces only, labeled, lengths that balance or don't.
- **Two problems**: diagram, $a = F_{net}/m$, forces with signs, letters first, units, sig figs.
- **Linearize data**: axes from $y = mx + b$, a ruler line, slope from points on your line.
- Not on it: forces at angles (components, inclines).

Free-body diagrams



at rest, $a = 0$ | speeding up, going up (up arrow longer) | sliding right, slowing (friction alone, left)

- F_g straight down, always. F_N perpendicular to the surface. F_f along it, opposite the sliding.
- F_T along the rope, away from the object. F_D opposite the motion. F_L across the air flow. F_{th} forward.
- No components, no net-force arrow, keep the picture's angles.
- $a = 0$: arrows balance. Otherwise the longer side points the way it accelerates.

Every problem, same chain

$a = \frac{F_{net}}{m} \rightarrow F_{net} = ma \rightarrow F_N - F_g = ma \rightarrow F_N - mg = ma$

$\rightarrow F_N = m(g + a) \rightarrow$ plug in with units $\rightarrow$ third law last

- Choose a positive direction; each force gets its sign from the diagram.
- $F_N > mg$ when accelerating up (speeding up going up or slowing going down).

Jumps: two stages

- Air stage (only F_g): $v_1 = \sqrt{2gh}$.
- Push stage (on the ground): $a_1 = \frac{v_1}{t}$, then $F_N = m(g + a_1)$.
- Third law: force on the ground = F_N, pointed down.

Rockets

- $F_{th} - mg = ma$, so $a = \frac{F_{th}}{m} - g$.
- Use the current mass. Fuel burned = $M - \frac{F_{th}}{g + a}$.
- Same thrust, less mass: the acceleration grows.

Linearizing

Plot	Physics equation	Slope means
a vs. 1/m	$a = F_{net}(1/m)$	F_{net}
x vs. t^2	$x = \frac{1}{2}a\,t^2$	$\frac{1}{2}a$, so $a = 2 \times$ slope
v^2 vs. x	$v^2 = 2a\,x$	$2a$
v^2 vs. 1/m	$v^2 = 2Fd(1/m)$	$2Fd$

- Fill the new column, label both axes with units, scale so data uses over half the grid.
- Best-fit line with a ruler, not forced through points or the origin.
- Slope from two far-apart points on the line, not data points; slope keeps its units.

Newton's laws

- 1st: $F_{net} = 0$ means constant velocity (at rest is one case).
- 2nd: $a = \frac{F_{net}}{m}$, a and F_{net} point the same way. $F_g = mg$, $g = 9.8$ N/kg.
- 3rd: $F_{A \text{ on } B} = -F_{B \text{ on } A}$. Same size, opposite way, different objects.

Traps

- Net force follows the acceleration, not the velocity.
- Third-law pairs act on different objects and never cancel.
- Grams to kg, ms to s, units on every number, sig figs from the givens.
- Weight is mg in newtons; mass is in kg.

Key: F_N normal force · F_g gravity (weight) · F_f friction · F_T tension · F_D drag · F_L lift · F_{th} thrust · F_{net} net force (ΣF) · Mg megagram (1000 kg)